exploring black holes.

Within days after formulating his equivalence principle, Einstein used it to make an amazing prediction, called *gravitational time dilation: If one is at rest relative to a gravitating body, then the nearer one is to the body, the more slowly one's time must flow.* For example, in a room on Earth, time must flow more slowly near the floor than near the ceiling. This Earthly difference turns out to be so minuscule, however (only 3 parts in 10^{16}; that is, 300 parts in a billion billion), that it is exceedingly difficult to detect. By contrast (as we shall see in the next chapter), near a black hole gravitational time dilation is enormous: If the hole weighs 10 times as much as the Sun, then time will flow 6 million times more slowly at 1 centimeter height above the hole's horizon than far from its horizon; and right at the horizon, the flow of time will be completely stopped. (Imagine the possibilities for time travel: If you descend to just above a black hole's horizon, hover there for one year of near-horizon time flow, and then return to Earth, you will find that during that one year of your time, millions of years have flown past on Earth!)

Einstein discovered gravitational time dilation by a somewhat complicated argument, but later he produced a simple and elegant demonstration of it, one that illustrates beautifully his methods of physical reasoning. That demonstration is presented in Box 2.4, and the *Doppler shift* of light, on which it relies, is explained in Box 2.3.

When starting to write his 1907 review article, Einstein expected it to describe relativity in a universe without gravity. However, while writing, he had discovered three clues to the mystery of how gravity might mesh with his relativity laws—the equivalence principle, gravitational time dilation, and the extension of his principle of relativity to include gravity—so he incorporated those clues into his article. Then, around the beginning of December, he mailed the article off to the editor of the *Jahrbuch der Radioaktivität und Elektronik* and turned his attention full force to the challenge of devising a complete, relativistic description of gravity.